Chem120 Midterm 1 Practice Exam 2

General Chemistry 2 (McGill University)

Practice Exam Midterm 1 (Chapter 4, 14, and 5)

Short Answers

Chem 120

Total Points: 75

Q1 (12 points)

a) **3 points**
Calculate the height of a column of liquid mercury (density = 13.7 g/cm^3) required to exert the same pressure as a 12 m column of water (density = 1.00 g/cm^3).

b) **3 points**
The partial pressures of CH_4, N_2, and O_2 in a sample of gas were found to be 155, 435, and 122 mmHg respectively. Calculate the mole fraction of each component.

c) **6 points**
A 1.325 g sample of an unknown vapor occupied 368 mL at 114 ˚C and 946 mm HG. Test show that the sample contains only oxygen and nitrogen in a ration N:O = 1:2. What is the molecular formula of the compound? Show complete calculations.

Q2 (9 points)

a) **3 points**
A certain amount of gas at 25 ˚C and at pressure of 0.800 atm is contained in a glass vessel. Suppose that the vessel can withstand a pressure of 2.00 atm. How high can you raise the temperature (˚C) of the gas without bursting the vessel?

b) **4 points**
Calculate the density of SF_6 gas at 27 ˚C and 0.500 atm pressure.

c) **2 points**
Plot the relationship between the number of moles of a gas (x axis) versus the volume of a gas (y axis), under constant pressure and temperature

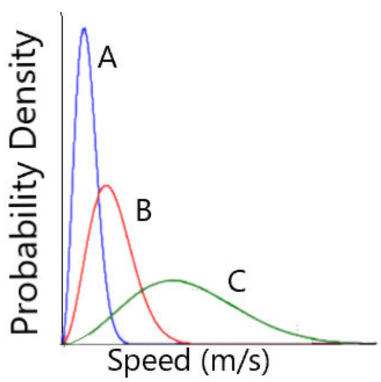
Q3) 8 points

a) 5 points

Calculate u_{rms} for CO_2 (g) molecules at 10 ˚C?

b) 3 points

A, B, and C are the Maxwell-Boltzmann distribution of speed plots for the same gas. Answer the following questions:



i. Gas with the highest u_{mp} (most probable speed): A or B or C

ii. Gas with the highest average kinetic energy: A or B or C

iii. What is most likely the difference between the experimental setup (conditions) between A, B, and C

Q4) 7 points

a) **5 points**

Following is an elementary reaction:

$$A \longrightarrow \text{Products}$$

i. What is unit of rate for this reaction?

ii. Write the rate law for this reaction?

iii. What is the unit of rate constant (k) for this reaction?

iv. Plot the rate of the reaction (y axis) as a function of concentration of A (denoted by [A] on x axis)

b) **2 points**

$$2NO(g) + 2H_2(g) \longrightarrow N_2(g) + 2H_2O(g)$$

Express the reaction rate for the above reaction in terms of : i. $(\Delta[NO])/\Delta t$ and ii. $(\Delta[N_2])/\Delta t$

i. Rate =

ii. Rate =

Q5) 10 points

a) **6 points**
For the reaction

$$2A + B \rightarrow \text{Products}$$

the following initial rate data were collected at constant temperature. Determine the correct rate law for this reaction (including the value of the rate constant).

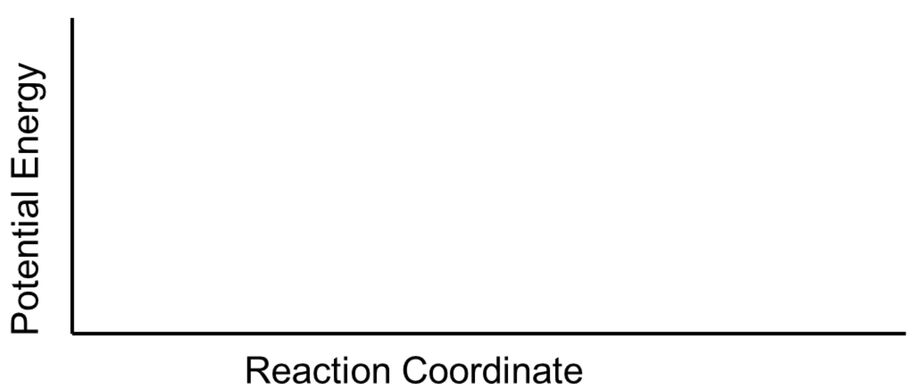
Trial	[A] (M)	[B] (M)	Rate (Ms^{-1})
1	0.150	0.100	0.022
2	0.300	0.100	0.044
3	0.300	0.200	0.022

b) **4 points**
The rate constant for the reaction A → products is 6.00×10^{-2} min^{-1}. How much time (in minutes) will it take the concentration of A to drop from 0.75 mol L^{-1} to 0.25 mol L^{-1}.

Q6) 10 points

a) **4 points**
In a first order reaction, $A \longrightarrow \text{Products}$, [A] = 0.500 M initially and 0.350 M after 12.0 min, at what time (in minutes) will [A] = 0.150 M?

b) **6 points**
For a one-step reaction, the activation energy of the forward reaction is 350 kJ/mol, and the activation energy for the reverse reaction is 150 kJ/mol. On the axes below, draw a reaction coordinate diagram for this reaction, showing the curve, and labeling i) ΔH ii) Forward activation energy, and iii) Reverse activation energy. State whether the reaction is endothermic or exothermic.



Q6) 9 points

a) **5 points**

Consider the following reaction:

$ClF(g) + F_2(g) \rightarrow ClF_3(g) \quad \Delta_rH° = ?$

Calculate the $\Delta_rH°$ for the reaction in kJ, using the information below:

$2ClF(g) + O_2(g) \rightarrow Cl_2O(g) + F_2O(g) \quad \Delta H = 167.4$ kJ

$2ClF_3(g) + 2O_2(g) \rightarrow Cl_2O(g) + 3F_2O(g) \quad \Delta H = 341.4$ kJ

$2F_2(g) + O_2(g) \rightarrow 2F_2O(g) \quad \Delta H = -43.4$ kJ

b) **4 points**

Calculate q, the heat change (in kJ) when 28.6 g of water is heated from 22.0°C to 78.3°C. $C_{water} = 4.186$ J/g°C

Q7) 10 points

a) 6 points

A piece of copper metal is initially at 100.0°C. It is dropped into a coffee cup calorimeter containing 50.0 g of water at a temperature of 20.0°C. After stirring, the final temperature of both copper and water is 25.0°C. Assuming no heat losses, and that the specific heat (capacity) of water is 4.18 J/(g·K), what is the heat capacity of the copper in J/K?

b) 2 points

Write down the formation reaction of ethanol (C_2H_5OH).

c) 2 points

A system delivers 225 J of heat to the surroundings while delivering 645 J of work. Calculate the change in the internal energy, ΔU, of the system